feb. 12

$\left\{\begin{array}{l}\text{all multiples of } a \\ \text{identity} \\ \text{all multiples of } a^{-1}\end{array}\right\}$ ← a^n, $n \in \mathbb{Z}^+$ ← identity ← a^m, $m \in \mathbb{Z}^-$

- Recall, a group/subgroup G is cyclic if $G = \langle a \rangle = \{a^n : n \in \mathbb{Z}\}$ = "the smallest subgroup containing a"
  - for a whole group, we say G is generated by a $(G,*)$ a^n is notation for $\underbrace{a*a*\ldots*a}_{n \text{ times}}$

Warm-ups/examples - determine whether or not the following are cyclic

- $\mathbb{Z}/4\mathbb{Z} = \{\bar{0}, \bar{1}, \bar{2}, \bar{3}\}$
  $\langle \bar{0} \rangle = \{\bar{0}\}$ ← trivial group or subgroup ×
  $\langle \bar{1} \rangle = \{\bar{1}, \bar{2}, \bar{3}, \bar{4} = \bar{0}, \}$ cycles back
  Cyclic

- $\mathbb{Z}/7\mathbb{Z}^\times = \{\bar{1}, \bar{2}, \bar{3}, \bar{4}, \bar{5}, \bar{6}\}$ ($\bar{6} = \bar{-1}$)
  $\langle \bar{1} \rangle = \{\bar{1}\}$ × trivial group
  $\langle \bar{2} \rangle = \{\bar{2}, \bar{4}, \bar{1}, \}$ ×
  $|\langle \bar{2} \rangle| = |\bar{2}| = 3$, but $|\mathbb{Z}/7\mathbb{Z}^\times| = 6$
  $\langle \bar{3} \rangle = \{\bar{3}, \bar{2}, \bar{-1}, \bar{4}, \bar{5}, \bar{1}\}$ ($\bar{-1} = \bar{6}$)
  $|\langle \bar{3} \rangle| = |\bar{3}| = 6 = |\mathbb{Z}/7\mathbb{Z}^\times|$, so
  Cyclic

- $\mathbb{Z}$ under $+$
  $\langle 0 \rangle = \{0\}$ ×
  $\langle 1 \rangle = \{\ldots -3, -2, -1, 0, 1, 2, 3 \ldots\}$
  incl. -1 since $1^{-1} = -1$, also generated by -1
  Cyclic

- $\mathbb{Z}/8\mathbb{Z}^\times = \{\bar{1}, \bar{3}, \bar{5}, \bar{7}\}$ ($\bar{7} = \bar{-1}$)
  Not cyclic
  every element has order 1 (id) or 2

- $D_3 = \langle r, s : r^3 = s^2 = e, rs = sr^2 \rangle$
  r^n cannot generate s ×
  s^n cannot generate r ×
  id cannot generate s, r ×
  Not cyclic

- $5\mathbb{Z} = \langle -5 \rangle = \langle 5 \rangle$
  Cyclic

subgroups of D_3: $\langle s \rangle, \langle rs \rangle, \langle r^2 s \rangle, \langle r \rangle$ — D_3 "improper"; id trivial; Proper subgroups

- You can check that the order of an element = order of group to check if the element generates the group (i.e it's cyclic)

- To be cyclic, the group needs only one element to be a cyclic generator
  - Can have more, i.e $\mathbb{Z}/7\mathbb{Z}$ has 2

$\langle \bar{2} \rangle < \mathbb{Z}/7\mathbb{Z}^\times$

$\left\{\begin{array}{l}\bar{2}, \bar{4}, \bar{1} \\ \bar{1} \\ \bar{4}, \bar{2}, 1\end{array}\right.$ $\quad \bar{2} \cdot x = \bar{1} \Rightarrow x = \bar{4}$

$\left\{\begin{array}{l}\text{all multiples of } a \\ \text{identity} \\ \text{all multiples of } a^{-1}\end{array}\right\}$ ← a^n, $n \in \mathbb{Z}^+$ ← identity ← a^m, $m \in \mathbb{Z}^-$

- Thm. 3.3 (The finite subgroup test) Let H be a nonempty, finite subset of a group G. If H is closed under the operation of G, then $H \leq G$.
  Proof We will use the 2-step subgroup test.
  Since H is closed, we need only prove that inverses are in H. In other words,

if $h \in H$, then $h^{-1} \in H$.

Let $a \in H$. Since H is closed, $a^n \in H$ for all $n \in \mathbb{Z}^+$. Since H is finite, we have $a^j = a^k$ for some $j, k \in \mathbb{Z}^+$ with $j \neq k$ (eventually have to cycle back). Without loss of generality, assume $j < k$, so letting $k = j + m$, $a^j = a^k = a^{j+m} = a^j \cdot a^m$. Thus in H and in G we have $a^j = a^j \cdot a^m$.

In G, we group operation on the left by $a^{-j} \Rightarrow a^{j-j} = e = a^{-j} \cdot a^j \cdot a^m = a^m$. Thus $a^m = e$. Since $j < k$ and $k = j + m$, we see $m \in \mathbb{Z}^+$, so $a^m \in H$. Thus $e \in H$. We continue with $a^m = e$.

We want $a^m = a \cdot a^{m-1} = e$, so $a^{-1} = a^{m-1}$ by uniqueness of inverse.

If $m > 1$, then $m - 1 \in \mathbb{Z}^+$, so $a^{-1} = a^{m-1} \in H$. ✓

if $m = 1$, them $a^m = a^1 = a = e$, and $e \cdot e = e$, so $a^{-1} = e \in H$. ✓ $\square$